Module 5 Assignment

Research gene: CYP2C19

1. Thresholds and Calibration

In this analysis, I selected CYP2C19 as the research gene, and set the maximum allele frequency threshold (MAX_AF) at 0.25. I first chose 0.20 as MAX_AF, but after seeing the results, I found it got rid of alleles related to drug metabolism function, such as CYP2C19*17, so I adjusted the allele-frequency threshold from 0.20 to 0.25.

This analysis initially obtained 80 CYP2C19 mutation records. After using MAX_AF= 0.25, 2 records were excluded and 78 records were kept. As a comparison, if MAX_AF = 0.001 is used, 19 records will be excluded, including rs4244285 (AF = 0.167953) and rs12248560 (AF = 0.200120). This result shows that overly strict low-frequency screening may omit mutations related to drug metabolism.

In the calibration part of the prediction tool, I chose the CADD PHRED score because it has the highest coverage rate in this data. The coverage rate of CADD is 85% (68/80), while both REVEL and AlphaMissense are 67.5%. I divide the CADD score into five intervals: (0,10], (10,20], (20,25], (25,30] and (30,50], and try to calculate the likelihood ratio (LR) of each interval. However, the reference data used for calibration has only 5 benign or potentially benign variants (B/LB), and there are no pathogenic or potentially pathogenic variants (P/LP). All 5 B/LB records are located in the (0,10] interval, and LR = 0.0 is calculated; the other four intervals do not have enough reference data, and LR is undefined.

Although the calibration program outputs LR = 0.0 in the (0,10] interval of CADD, this value cannot be regarded as reliable benign evidence. The dataset contained only five B/LB reference variants and no P/LP reference variants, so meaningful **likelihood ratio**s could not be estimated. At the same time, the sample size of each interval is less than 10, which further limits the reliability of the calibration results.

Therefore, I use the small-gene route to make it clear that the current data is not enough for reliable calibration, instead of forcibly classifying according to the CADD score. In the end, all 33 uncertain mutations (VUS) were retained as VUS because the existing evidence was insufficient to support reclassification.

2. Coverage gaps and the regulatory genome.

This analysis found that there are obvious differences in the coverage of variant types by different prediction tools. The overall coverage rate of CADD is 85%, while both REVEL and AlphaMissense are 67.5%. REVEL does not provide scores in seven categories of mutation consequences, including intergenic_region, intron_variant, splice_donor_variant, stop_gained, stop_lost, synonymous_variant and upstream_gene_variant. Since REVEL is mainly used for missense variation prediction, these missing values reflect it do not mean that the relevant variation has no functional impact.

Rs4244285 is a representative example. It is marked as synonymous_variant in this data. Both REVEL and AlphaMissense do not provide scores, while CADD PHRED is 5.686. This shows that relying only on the missense variation prediction tool may not be able to fully evaluate other functional effects.

In order to further explore these coverage blind areas, I used AlphaGenome Atlas and obtained a total of 70 AVI scores of variation records, and no query failure was reported. For rs4244285, AVI is 22.33, and the main attribution feature is MERGED_SPLICING. According to the course materials, this feature is related to RNA splicing, so this result suggests that rs4244285 may have a functional impact related to splicing.

The prediction results of other mutations also show different attribution characteristics. For example, the AVI of rs4986893 is 32.43, and the top_feature is PROTEIN_TERMINATION; the AVI of rs72558186 is 25.51, and the top_feature is MERGED_SPLICING; The AVI of rs55640102 is 25.48, and the top_feature is STOP_LOST. Among the 70 variants with AVI scores, the most common main feature is ALPHAMISSENSE (40), followed by CACTUS_241_WAY (15) and MERGED_SPLICING (6). These features provide different types of evidence. CACTUS_241_WAY reflects conservation but does not establish a specific molecular mechanism, while MERGED_SPLICING suggests a possible splicing-related effect that could be tested using an RNA assay.

For rs4244285, I think the next step is to use RNA experiments to compare whether there are different CYP2C19 RNA splicing products in samples with and without mutations. This will help to judge whether the splicing mechanism proposed by AlphaGenome really exists.

3. A discordant variant, read mechanistically

In this analysis, I chose rs145328984 (R73C) as the main research object. The variation belongs to missense_variant and triggers the flag_cadd_vs_revel rule because its CADD PHRED score is greater than 25 (26.0) and the REVEL score is below 0.35 (0.125).

According to the results of this analysis, the CADD PHRED of rs145328984 is 26.0, REVEL is 0.125, AlphaMissense is 0.2085, and AlphaGenome AVI is 11.27. These results show that different prediction tools give different degrees of functional impact signals to the mutation. CADD showed a higher prediction score, while REVEL and AlphaMissense did not show the same strong signal of protein function damage.

I think this difference may be related to the forecast information that different tools pay attention to. CADD synthesizes multiple genome annotation information, while REVEL and AlphaMissense are mainly used to assess the potential impact of missense variation on protein function. Therefore, a position may have characteristics that make the CADD score higher, but the specific amino acid replacement that occurs in that position does not necessarily cause serious functional damage.

R73C means that arginine (R) at position 73 of the CYP2C19 protein is replaced by cysteine (C). Although the chemical properties of the two amino acids are different, this replacement alone cannot determine whether it will significantly affect the function of the protein.

Combined with the existing prediction results, I am more inclined to believe that rs145328984 may be located in an area with certain sequence constraints, but the specific replacement of R73C may not cause serious damage to protein function. This explanation can illustrate the higher CADD and the lower results of REVEL and AlphaMissense, and is also consistent with the discussion of the differences between cadd_vs_revel rule.

However, this analysis did not directly measure the protein function or enzyme activity of CYP2C19, so the real functional impact of R73C could not be determined. I believe further functional experimental evidence is still needed.

4. Limitations and Next Experiment

Although multiple computational prediction tools are used in this analysis, I still can't identify two important problems for rs145328984 (R73C).

First, I can't determine the actual impact of the mutation on the activity of CYP2C19 enzyme. The existing prediction score cannot directly show that R73C will increase or decrease, or basically do not change the enzymatic activity of CYP2C19, nor can it determine its impact on the metabolic efficiency of specific drugs.

Second, I can't judge the actual drug metabolism phenotype of the patient based on the prediction results of a single mutation alone. The patient's complete genotype, heterozygous or homozygous status, allele phase, and other drug factors may affect the final drug

metabolism. Therefore, the existing data is not enough to predict the response of real patients to drugs.

If I choose one single experiment, I will use CYP2C19 in vitro enzyme activity experiment. Under the same conditions, the metabolic activity of wild CYP2C19 and R73C variants on the same drug substrate was compared, and the protein expression level was measured at the same time to help distinguish between expression changes and enzyme function changes.

If the metabolic activity of R73C is similar to the wild type, I will support the explanation in the third question, which is that the mutation may not cause obvious functional damage. In contrast, if the experiment shows that the metabolic activity of R73C declines, I need to reconsider this explanation and further study whether the existing prediction tools underestimate the functional impact of the variation.